
Heterogeneous Preferences and the Efficacy of Public School Choice

by Hastings, Kane, and Staiger

ECON 649

Basic Discrete Choice Framework

$$u_{ij} = x_j \beta + \varepsilon_{ijt}$$

$$u_{ij} = \alpha(y_i - p_j) + x_j \beta + \varepsilon_{ijt}$$

$$u_{ijt} = \alpha_i(y_i - p_{jt}) + x_{jt} \beta_i + \xi_{jt} + \varepsilon_{ijt}$$

- If one of the product characteristics is price (or expenditures), can use to convert utils to \$ and do Consumer Surplus/Welfare calculations
- Can allow heterogeneity in preference parameters
- Estimation via:
 - MLE if no “endogeneity” (See Train)
 - GMM if need to IV for e.g. price (See BLP or Nevo)
- If you prefer the movie over the book, see NBER Summer Institute methods lecture by Guido Imbens: <http://vimeo.com/27776865>

Why Allow School Choice

- Opportunities for parents to choose schools that better match their preferences (e.g. academics, demographics of other kids)
- Demand-side pressure: schools incentivized to improve if parents/kids leaving
- Will the competitive tide “life all boats”?
- Main insight here: could be substantial heterogeneities in preferences
 - Some parents more elastic w.r.t. academics, same-race
 - To analyze, need a model that allows for heterog
 - Surprising finding: can increase racial score gap, and actually lower the academic performance of African American kids

Introduction of Choice to Charlotte-Mecklenburg Public Schools

- Prior to 2002, operated under a order to racially desegregate; lots of busing, sometimes long distances; court declared “unitary”
- Fall 2002: school choice to be implemented
- New school boundaries drawn; “home-school” assigned
 - Very different boundaries -- 43% of elementary students had a new “home-school”
- Spring 2002: parents could submit top 3 choices
 - 105K of 110K students submitted “choices”
 - If a school was oversubscribed, rationing by lottery

Product Characteristics

- Academic “quality” of school
- Distance from student’s home to school
- Own-race: might create larger tradeoff for minorities (in neighborhoods with weaker schools)

Table I: Descriptive Statistics for Students, Schools, and Parental Choices

	Not Receiving Lunch Subsidies		Receiving Lunch Subsidies	
	White	Not White	White	Not White
<i>Student Characteristics</i>				
Baseline test score				
Neighborhood Income				
<i>Student-School Characteristics</i>				
Driving Distance to the 'Home-School'				
Driving Distance to Closest 'non-Home-School'				
Home School Ave. Test Score				
Range of School Test Scores in 6 miles				
Highest Scoring School in 6 miles				
<i>Choice Characteristics</i>				
Chose Home School				
Submitted 1 Choice				
Submitted 2 Choices				
Submitted 3 Choices				
Observations				

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<i>Student Characteristics</i>				
Baseline test score	0.638 (0.825)	-0.091 (0.839)	-0.089 (0.849)	-0.613 (0.799)
Neighborhood Income	\$73,804 (25,869)	\$50,612 (21,502)	\$52,763 (22,556)	\$36,455 (16,230)
<i>Student-School Characteristics</i>				
Driving Distance to the 'Home-School'	3.298 (2.085)	2.915 (1.777)	2.725 (1.759)	2.445 (1.625)
Driving Distance to Closest 'non-Home-School'	3.752 (2.008)	3.091 (1.765)	3.004 (1.808)	2.269 (1.393)
Home School Ave. Test Score	0.239 (0.384)	-0.160 (0.368)	-0.147 (0.377)	-0.385 (0.357)
Range of School Test Scores in 6 miles	0.670 (0.503)	0.817 (0.438)	0.827 (0.448)	1.018 (0.367)
Highest Scoring School in 6 miles	0.530 (0.342)	0.288 (0.317)	0.317 (0.329)	0.314 (0.268)
<i>Choice Characteristics</i>				
Chose Home School	0.647	0.405	0.529	0.361
Submitted 1 Choice	0.512	0.254	0.321	0.181
Submitted 2 Choices	0.202	0.186	0.203	0.149
Submitted 3 Choices	0.286	0.559	0.475	0.670
Observations	16,187	6,194	2,134	12,372

Whites:
Better
academic
home
schools

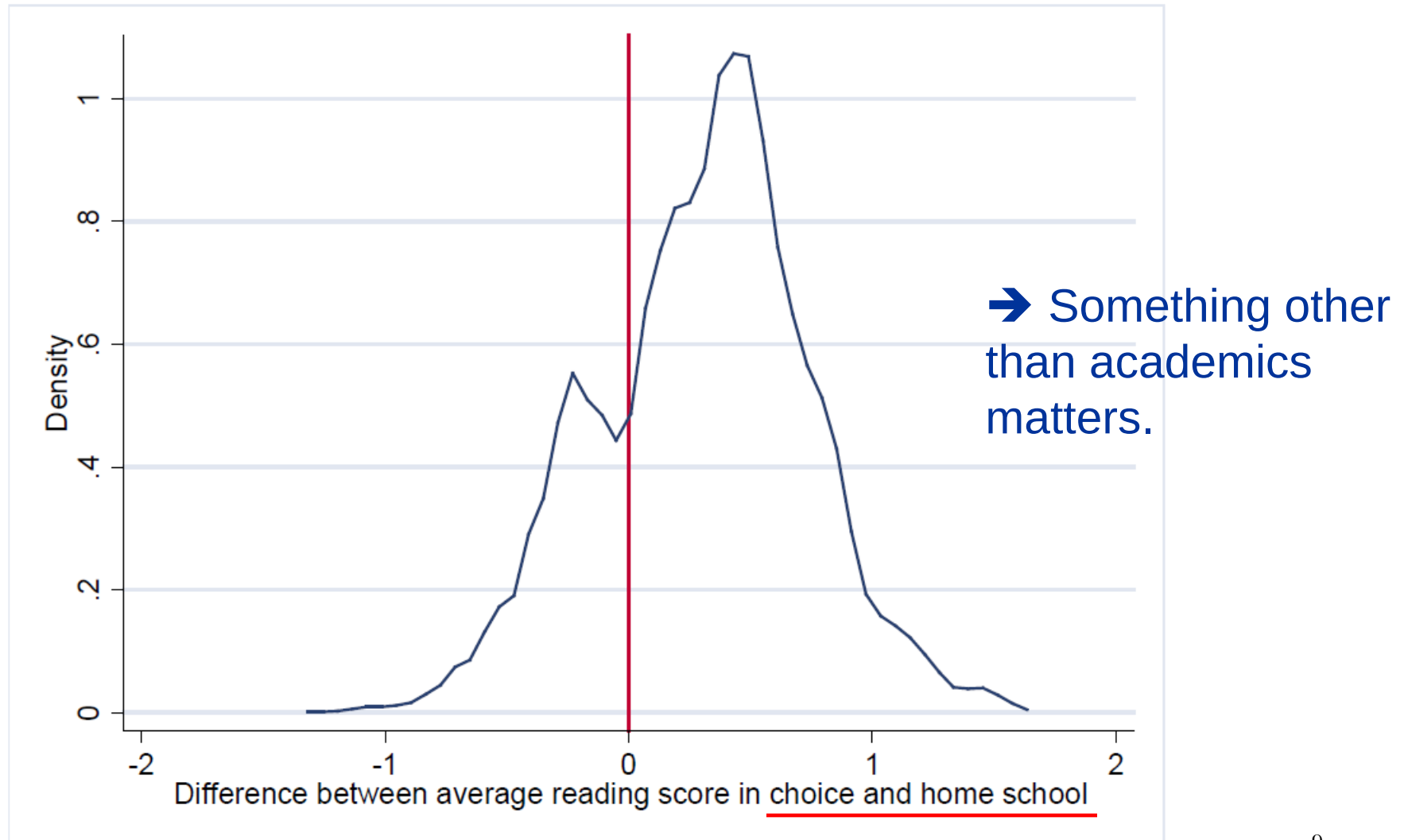
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Better
nearby
schools
available
for
everyone

School Isn't a Homogeneous Good...

Figure 1: Distribution of Difference in Average Standardized School Score Between Student's First-Choice School and Home School



A School as a Product with Characteristics

- Academics
 - Proxied by avg test score (S_j)
 - Allow for student to have idiosyncratic “match”
- Distance to school
- Racial composition

$$U_{ij} = \beta_i^A \kappa_i (S_j + \gamma_{ij}) + X_{ij} \beta_i^X + \omega_{ij}$$
$$= S_j \beta_i^S + X_{ij} \beta_i^X + \varepsilon_{ij} \quad \text{where } \beta_i^S = \beta_i^A \kappa_i \text{ and } \varepsilon_{ij} = \beta_i^S \gamma_{ij} + \omega_{ij}$$

- High weight on test score due to large parent preference (β_i^A) or child sensitive to school (κ_i)
- Caveat: β_i^A also picks up information / decision costs

Will Choice Improve Test Scores (compared to home-school)?

- Not necessarily!
- Extreme:
 - parents place no weight on academics ($\beta_i^A=0$) and positive weight on “other” (e.g. same-race) ($\beta_i^X>0$)
 - If academics and “other” are negatively correlated, optimal choice will yield school with lower academics

$$U_{ij} = \beta_i^A \kappa_i (S_j + \gamma_{ij}) + X_{ij} \beta_i^X + \omega_{ij}$$

- Expected treatment effect more likely positive if:
 - Parents place sufficient weight on academics relative to other ($\beta_i^A>0$)
 - Student sensitive to school quality ($\kappa_i>0$)
 - Academics and “other characteristics” not strongly negatively correlated

Data to Exploit

- Parents' listed top 3 choices
 - Even more information that “typical” discrete choice where actual choice indicated 1st choice
 - S_j = 2002 average test scores in school j
 - X_{ij} = driving distance to school j for student i , %African American (and its square), indicator of whether in Choice Zone (free busing), indicator of previous year school (inertia), indicator of student's home-school
 - Use “*exploded mixed logit*”
- Lottery assignments to estimate impact of attending first choice school

Does Order of Choices Reflect the Rank Order of Preferences?

- Concern is rationing may encourage parents to “play it safe” by listing a less preferred school with a perceived higher chance of getting in
- Authors claim such strategic behavior unlikely
 - Parents didn’t know rationing mechanism and district created impression that they’d try to get student into one of chosen schools
 - District gave “priority boost” to low-income students in schools with fewer low-income students → less incentive to hedge by low income parents

“Exploded” Logit

- Suppose 5 schools in choice set

- Consumer i :

- 1st: school #1
- 2nd: school #2
- 3rd: school #3



- $U_{i1} \geq U_{ik}$ for $k=2,3,4,5$ } “Standard” logit
- $U_{i2} \geq U_{il}$ for $l=3,4,5$ } Provide additional information on how
- $U_{i3} \geq U_{im}$ for $m=4,5$ } consumer trades off product characteristics

“Mixed” Logit

- Allow preference parameters to be random variables
- Generally, parameterize the random variable
 - $\beta \sim f(\beta | \mu_\beta, \theta)$
- Simplest: Just allow var-cov
- More general: parameterize to consumer characteristics
 - E.g. Nevo’s cereal approach

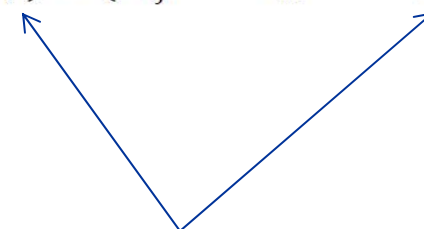
Likelihood Function for Exploded Mixed Logit

$$\begin{aligned}
 P_i(j^1, j^2, j^3) &= Pr\{(U_{ij^1} > U_{ik} \forall k \in J_i^1) \cap (U_{ij^2} > U_{ik} \forall k \in J_i^2) \cap (U_{ij^3} > U_{ik} \forall k \in J_i^3) \cap \} \\
 (7) \quad &= \int \prod_{c=1}^3 \frac{\exp(S_j \beta_i^S + X_{ij} \beta_i^X)}{\sum_{k \in J_i^c} \exp(S_k \beta_i^S + X_{ik} \beta_i^X)} f(\beta | \mu, \theta) d\beta
 \end{aligned}$$

These probabilities form the log-likelihood function: ¹²

$$(8) \quad LL(X, \mu, \theta) = \sum_{i=1}^N \sum_{j=1}^{J_1} \sum_{k=1}^{J_2} \sum_{l=1}^{J_3} d_{ij}^1 d_{ik}^2 d_{il}^3 \ln(P_i(j, k, l))$$

Choice set when exclude goods already chosen



- No closed-form solution, so simulate LL by taking 300 draws and integrating numerically

Parameterization of Utility

- Estimate separately for all 4 combos of: white/non-white & receive lunch subsidy/not → “fully interacted”
- Parameterization of random coeffs:
 - Mean prefs (i.e. μ_β) vary by elementary or middle
 - Mean pref for academics (i.e. μ_β^A) vary in student’s prior test score, neighborhood income
 - Key: do not restrict covariance between idiosyncratic prefs for academics and proximity (distance & home-school); this allows for the “negative correlation”
 - NB: not sure why didn’t do same for same-race. Did I miss something?

Identification in structural models

- Very important slide!! When you present, spend lots of time here, especially with “A&Ps” in the audience
 - *Not* making an econometric point about unique parameter maximizing objective function!
 - Tell audience what in the data is making parameter estimate be whatever it is
 - Best case: becomes more transparent & believable
 - More modest goal: for those skeptical of your method, at least they'll know what they need to believe (even if they choose not to believe it)
- But causality police aside, convince yourself!!!
 - NEVER estimate a model before you can convincingly explain identification to yourself! Otherwise, we deserve to be criticized!

Identification in this paper (my understanding)

- Do not want idiosyncratic tastes for schools to be correlated with “product characteristics” (like ξ_j and p_j)
 - E.g. distance. Tiebout sorting could create this
 - Response: new school assignment zones very different from the effective old zones. Also, exploit 2nd choice data.
 - Me: sounds like only mitigated endogeneity. They do robustness test using only students switched to a new zone.
- Variation:
 - Within short distance, most students had schools with large variety in scoring
 - Additional “choices” from “exploded” logit remove previous chosen school so provide more variation from which to infer tradeoffs. (See bottom p.17 for good explanation)
- 95% of parents submitted choices, so not huge selection issues

Estimating Demand Side Pressure for a School to Improve

- How will a decrease in test scores reduce enrollment?

$$(9) \quad \frac{\partial Q_j}{\partial S_j} = \sum_{i=1}^N \frac{\partial \hat{P}_{ij}}{\partial S_j}$$

- Where P_{ij}^{hat} is the estimated probability that student i chooses school j

Data for AY 2001-02 & 2002-03

- All students grades 4-8 year before (for baseline score) and year after choice
- Reading & math test scores (normalized)
- Demographics
- Sample = 36,887 students

Table IV: Estimates from Exploded Mixed Logit Model of Demand for Schools

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Variable	Parameter	Not Receiving Lunch Subsidies		Lunch Subsidy Recipient	
		White	Not-White	White	Not-White
<i>Preferences for Test Scores</i>					
School Score	Mean Elementary				
	Mean Middle				
	St.Dev.				
Income* School Score	Mean Elementary				
	Mean Middle				
	St.Dev.				
Own Baseline Score*School Score	Mean Elementary				
	Mean Middle				
	St.Dev.				
<i>Preferences for Proximity</i>					
Distance	Mean Elementary				
	Mean Middle				
	St.Dev.				
Home School	Mean Elementary				
	Mean Middle				
	St.Dev.				
<i>Preferences for Race</i>					
Percent Black	Mean Elementary				
	Mean Middle				
	St.Dev.				
Percent Black Squared	Mean Elementary				
	Mean Middle				
	St.Dev.				
Implied Preferred % Black	Mean Elementary				
	Mean Middle				
	St.Dev.				
<i>Other Preferences</i>					
Last-Year's School	Mean Elementary				
	Mean Middle				
	St.Dev.				
In Choice Zone	Mean Elementary				
	Mean Middle				
	St.Dev.				
<i>Estimated Correlation Coefficients</i>					
Corr(Distance,School Score)					
Corr(Distance,Home School)					
Corr(School Score,Home School)					

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<i>Preferences for Test Scores</i>					
School Score	Mean Elementary	1.461	2.103	0.102	1.444
	Mean Middle	1.888	2.714	1.157	1.398
	St.Dev.	0.402	0.204	0.664	0.321
Income* School Score	Mean Elementary	0.017	0.014	0.004	0.010
	Mean Middle	0.015	0.013	0.010	0.004
	St.Dev.	--	--	--	--
Own Baseline Score*School Score	Mean Elementary	0.378	0.379	0.152	0.379
	Mean Middle	0.587	0.451	0.309	0.243
	St.Dev.	--	--	--	--
<i>Preferences for Proximity</i>					
Distance	Mean Elementary	-0.385	-0.335	-0.488	-0.357
	Mean Middle	-0.355	-0.245	-0.303	-0.235
	St.Dev.	0.083	0.070	0.183	0.096
Home School	Mean Elementary	2.110	1.801	1.804	1.709
	Mean Middle	2.150	1.659	1.953	1.714
	St.Dev.	0.826	0.650	0.314	0.259
<i>Preferences for Race</i>					
Percent Black	Mean Elementary	3.162	5.565	1.812	4.414
	Mean Middle	4.912	6.064	3.174	3.559
	St.Dev.	3.114	1.512	2.306	1.037
Percent Black Squared	Mean Elementary	-4.582	-2.666	-3.424	-2.032
	Mean Middle	-6.523	-3.977	-4.102	-2.856
	St.Dev.	--	--	--	--
Implied Preferred % Black	Mean Elementary	0.345	1.044	0.265	1.086
	Mean Middle	0.376	0.762	0.387	0.623
	St.Dev.	0.340	0.283	0.337	0.255
<i>Other Preferences</i>					
Last-Year's School	Mean Elementary	4.830	4.505	4.128	4.022
	Mean Middle	3.343	2.696	3.062	1.986
	St.Dev.	2.606	2.858	3.330	3.332
In Choice Zone	Mean Elementary	1.666	1.325	2.170	1.711
	Mean Middle	1.213	1.306	1.902	1.562
	St.Dev.	1.117	1.223	1.590	1.279
<i>Estimated Correlation Coefficients</i>					
Corr(Distance,School Score)		0.449	0.886	0.272	0.561
Corr(Distance,Home School)		0.107	0.136	0.281	0.341
Corr(School Score,Home School)		-0.215	-0.053	-0.668	-0.585

All groups prefer:

- Higher scores
- Closer school
- Majority same race
- Last year's school
- Busing
- Middle school parents place more weight on academics than non-academics (dist, race, last year, busing)

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Distance	Mean Elementary	-0.385	-0.335	-0.488	-0.357
	Mean Middle	-0.355	-0.245	-0.303	-0.235
	St.Dev.	0.083	0.070	0.183	0.096
Home School	Mean Elementary	2.110	1.801	1.804	1.709
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Percent Black Squared	Mean Elementary	-4.582	-2.666	-3.424	-2.032
	Mean Middle	-6.523	-3.977	-4.102	-2.856
	St.Dev.	--	--	--	--
Implied Preferred % Black	Mean Elementary	0.345	1.044	0.265	1.086
	Mean Middle	0.376	0.762	0.387	0.623
	St.Dev.	0.340	0.283	0.337	0.255
<i>Other Preferences</i>					
Last-Year's School	Mean Elementary	4.830	4.505	4.128	4.022
	Mean Middle	3.343	2.696	3.062	1.986
	St.Dev.	2.606	2.858	3.330	3.332
In Choice Zone	Mean Elementary	1.666	1.325	2.170	1.711
	Mean Middle	1.213	1.306	1.902	1.562
	St.Dev.	1.117	1.223	1.590	1.279
<i>Estimated Correlation Coefficients</i>					
Corr(Distance,School Score)		0.449	0.886	0.272	0.561
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Heterogeneity:

Weights on score bigger:

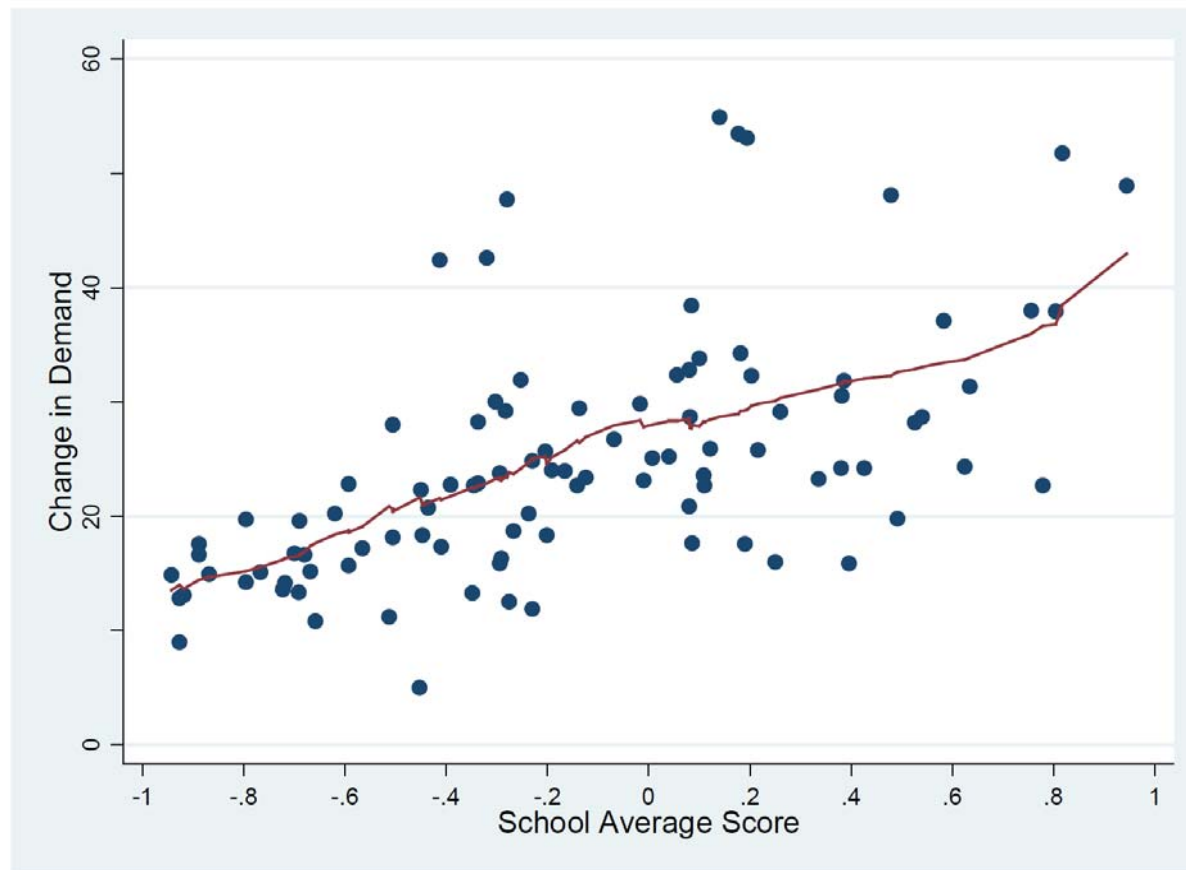
- Higher income
- Higher baseline score
- Non-whites
- Generally, lots of variation in score pref (see St.Dev.)

Does Choice Incentivize Schools?

- Model allows us to simulate choices when schools have different product characteristics
- Counterfactual simulation:
 - Each school (individually) given mean score 0.34 student-level std. deviations higher
- Questions:
 - How many more student choose it?
 - Are those students going to pull up or pull down school avg?
I.e. Will school induce switching by good or bad students?

All Schools Will Get More Students, Especially the Better Schools

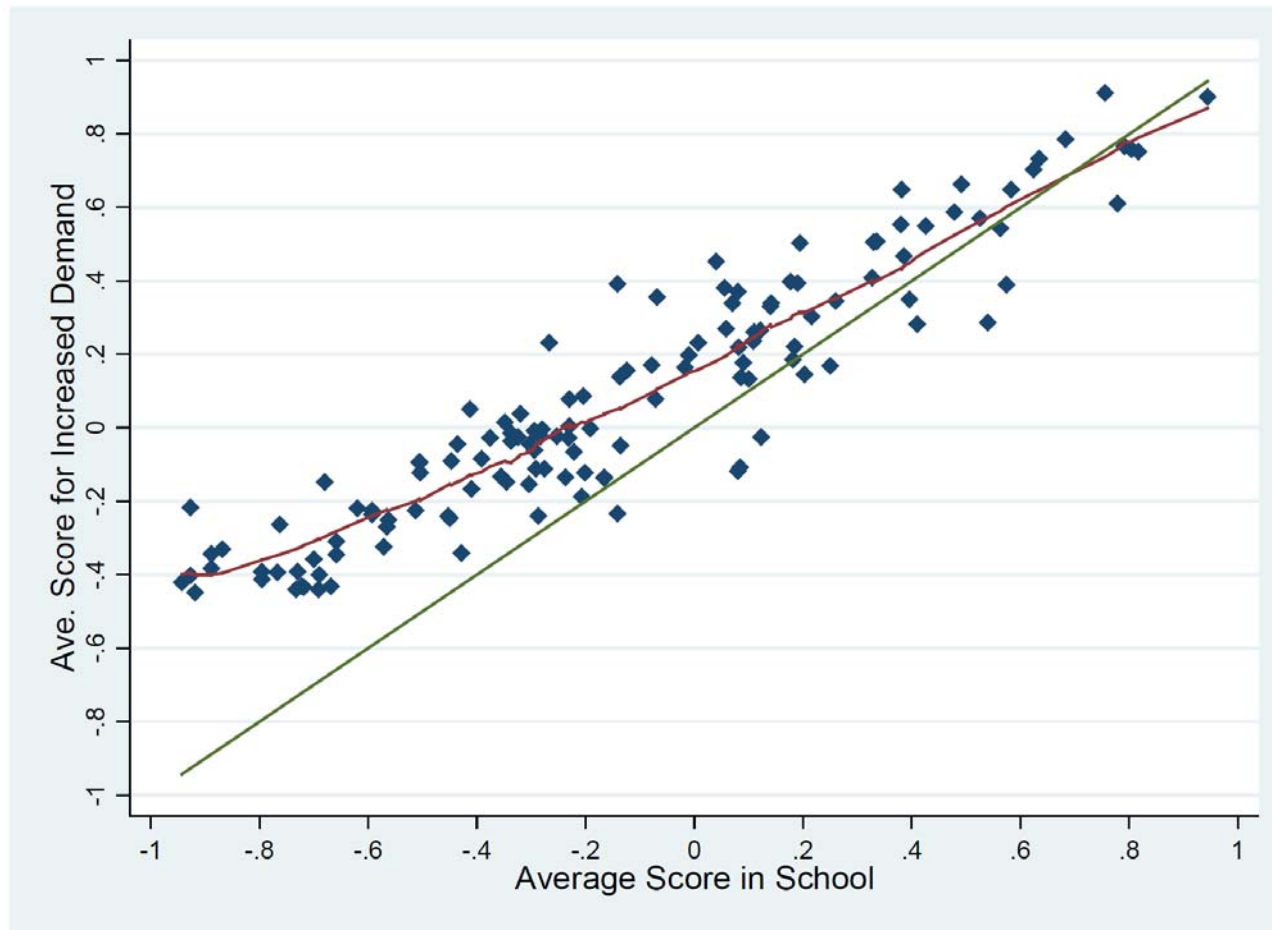
Figure 2a: **Elementary** Schools: Simulated Change in Number of Students Choosing School j When Average Standardized Score at School j Increases by 0.34 Points



Gradient even steeper at Middle Schools (expected given param estimates)

Students Switching Are Better than Average Students

Figure 3: Average 2002 Standard Deviation Scale Score for Additional Students Who Choose School j in Response to 0.34 Point Increase in Average Score at School j



Why Does Model Predict This?

- Weight on score larger for students with higher baseline score

Implications of Allowing School Choice on Test Score Gap

- Incentive to improve school quality strongest among high performing schools
- Switching students are high performing students
- So...choice likely to widen gap between low and high performing schools

Upshot

- Modeling perspective:
 - Heterogeneous treatment effects
 - Explaining effect with β_i^S allows to get at mechanism
- Policy Implications for School Choice:
 - Can widen “score gap” at school level
 - Can widen “score gap” at student level for high vs. low SES

The End

Estimating the Treatment-on-Treated Parameter

- Treatment effect of attending 1st choice school (vs. usually the home-school)
- Will use lottery as an IV for attending 1st choice

Using Lottery to Estimate Treatment Effect of Attending 1st Choice School

- Question: How is the treatment effect associated with parental preferences for academics (β_i^S)
- Sample: students choosing oversubscribed schools who were not given a special priority
- Lottery instrument seems valid
 - Randomization tests passed: winning not predicted by covariates, winning increases pr(attend 1st choice)

IV Results: Heterogeneity of Treatment Effect

Table VII: IV Estimates of Impact of Attending First-Choice School with Heterogeneous Treatment by Weight Placed on Academics in Choice Decision

<i>Dependent Variable:</i> <i>Combined Score in Spring 2003</i>	All Students		Students Who Prefer School Less Than 50% Black		Students Who Prefer School at Least 50% Black	
	(1)	(2)	(3)	(4)	(5)	(6)
Attended First-Choice School	-0.006 (0.049)	-0.176 (0.091)	0.135* (0.066)	0.252 (0.209)	-0.050 (0.061)	-0.230 (0.116)
$\hat{\beta}_i^S$ $\rightarrow$ <i>Weight</i> * attended First-Choice School		0.090** (0.032)		-0.050 (0.072)		0.103* (0.047)
P-value for Interaction with <i>Weight</i>		0.007		0.487		0.033
Joint P-Value on Reported Coefficients	0.907	0.012	0.047	0.105	0.415	0.101
Observations	2,591	2,591	720	720	1,871	1,871

Notes: Each column in the table is from a separate IV regression. The dependent variable is a student's combined standardized test score in the spring of 2003. Each specification reports the coefficients on attending the first-choice school and its interaction with the weight that the parent places on test scores ($Weight = \hat{\beta}_i^S$) in the school choice decision, using random assignment to the first-choice school and its interaction with *Weight* as instruments. All specifications control for lottery fixed effects, home school fixed effects, a direct control for the student's *Weight* estimate, and the student baseline covariates listed in the text. Sample includes only students in the randomized priority group with complete baseline data. Standard errors adjust for clustering at the level of the first-choice school. Asterisks indicate significance (*=.05, **=.01, ***=.001).

Model with no heterog

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→ Avg β_i^S – no effect
 95%ile β_i^S -- substantial effect (0.10)

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	(1)	(2)	(3)	(4)	(5)	(6)
Attended First-Choice School			0.135* (0.066)	0.252 (0.209)	-0.050 (0.061)	-0.230 (0.116)
<i>Weight</i> * attended First-Choice School				-0.050 (0.072)		0.103* (0.047)
P-value for Interaction with <i>Weight</i>				0.487		0.033
Joint P-Value on Reported Coefficients			0.047	0.105	0.415	0.101
Observations			720	720	1,871	1,871

- * Everybody prefers “own race”
- * $\text{Corr}(\text{school white}, \text{school score}) > 0$
- * So whites benefit regardless
- * Blacks benefit only if place strong weight on academics

➔ School choice can hurt academic achievement of students it is most intended to help (minorities from disadvantaged background)